

CRAYON: Supporting the Elicitation of Pluralistic Normative Requirements for Autonomous Systems

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I. EXTENDED ABSTRACT

Normative requirements capture social, legal, ethical, empathetic, and cultural (SLEEC) expectations that autonomous systems must satisfy when operating in human-centered environments. While regulatory frameworks such as the EU AI Act provide high-level principles, transforming these principles into concrete, system-specific requirements remains a challenging and highly multidisciplinary task. Existing approaches primarily focus on analyzing or verifying requirements once they have been specified, providing limited support for the elicitation process itself.

To address this challenge, we developed CRAYON, a human-in-the-loop, tool-supported framework for eliciting pluralistic normative requirements from high-level normative principles. CRAYON guides stakeholders through a structured workflow that links principles to system capabilities through intermediate artifacts, including proxies, capability mappings, concerns, and refinements. At each stage, LLM-assisted recommenders generate candidate artifacts that stakeholders can accept, modify, or reject, while preserving full traceability from principles to requirements.

We evaluated CRAYON with four multidisciplinary stakeholders (lawyer, philosopher, psychologist, and ethicist) on two real-world case studies: DAISY, an AI-assisted health-care triage system, and TherapyBot, a conversational system providing emotional support and therapeutic guidance. The evaluation showed that stakeholders accepted 86% of generated candidates and that CRAYON nearly doubled the number of elicited requirements compared with a prior manual elicitation process while increasing capability coverage and concern identification. These results suggest that structured, tool-supported elicitation can improve the completeness of normative requirements without constraining stakeholder creativity or disciplinary perspectives. In this talk, we will present the CRAYON workflow, discuss lessons learned from applying the framework in healthcare and therapy domains, and showcase a subset of the normative requirements elicited by multidisciplinary stakeholders. Example requirements include:

- When a user discloses thoughts of self-harm, then TherapyBot shall immediately provide crisis-support resources and encourage contact with a qualified professional.

- When TherapyBot provides therapeutic advice, then it shall clearly communicate that it is not a licensed mental-health professional.
- When a patient begins a new interaction with DAISY, then DAISY shall explain its capabilities and limitations in language understandable to the patient.
- When DAISY provides information that has not yet been verified by a clinician, then it shall provide a disclaimer indicating that the information may be inaccurate.
- When a user requests access to their personal information, then the system shall provide access only after verifying the user's identity.
- When sensitive personal information is collected, then the system shall store and process that information according to applicable privacy requirements.

These examples illustrate how abstract principles such as transparency, autonomy, safety, beneficence, and privacy can be operationalized into concrete, observable, and testable system requirements.